07/10/2025

Last edited time	October 28, 2025 11:13 AM
Subject	Server Scripting

Lesson Notes

- SSS-Lab Login Name: **sss**
- SSS-Lab Login Password: **password**

What is a Server?

- A computer, lol.
- The differences between Servers and Computers is that Servers are made to run 24/7, and has redudant storage in the case where one of the storage devices used by the server ever dies.

What is Server Side Scripting?

- This is a method of designing websites so that the process or user request is run on the originating server. Server-side scripts provide an interface to the user and limit access to proprietary data and help keep control of the script.
- Many people use Client Side Websites as most people do not have access of Servers, so they will use a provide to allow them to upload their own websites.

What is Server Side Rendering?

- The problem of websites built on client side frameworks (React, Angular and etc) is that they have undesirable SEO (Search Engine Optimisation).

> Through Server Side Rendering, search engine bots will be capable of seeing all the content regarding the website, making it search engine optimised.

Why is Laravel run on Virtual Machine instead on your laptop?

- Resource Efficency
- Operating System Efficency
- Portablility / Compatability
- Safety

In ths Unit, we will be using a virtual machine to pretend that a server is run locally for speed purposes.

How to find the IP of the server so you can use the server through command prompt on your device:

- Enter:

```
ip a
```

- “1: LO” is the Local Interface, this can be ignored.
- INET (IPv4) and INET6 (IPv6) are the addresses of the server.
- INET (IPv4) has the desired IP for the next step.
- If the INETs are empty, there are 2 possibilities:
  - The OS is not compatabile with the device the virtual machine is on
  - The device is not connected to the internet when running the virtual machine
- In this case, the IP is 172.16.233.128.
- Open command prompt on your device, and enter:

```
ssh sss@172.16.233.128
```

- Command prompt will then ask you for the password for User **sss**.
- After entering the password successfully, you will be able to access the virtual machine through the command prompt.

IMPORTANT NOTE, SHUT OFF THE VIRTUAL MACHINE BY PRESSING ON THE PAUSE ICON AT THE TOP LEFT, **DO NOT CLOSE THE VIRTUAL MACHINE WITH THE CLOSE BUTTON ON THE TOP RIGHT!!**

13/10/2025

Last edited time	October 28, 2025 11:13 AM
Subject	Server Scripting

Lesson Notes

DO NOT UPGRADE Laravel

To check what version of php you have, you can check by using the command

```
php -v
```

In this case, the virtual machine is using PHP 8.2.10

These are the exensions that must be downloaded on Visual Studio for this Unit:

- Remote - SSH
- Remote - SSH Editing Configurations Files
- Remote Explorer

This Unit consists of a device communicating with the Server (Which is on the VM), and with VS Code.

Now, how will VS Code communicate with the server? This is through the extensions that will help VS Code communicate with the server

Here are the steps:

- After installing the extensions, go to remote explorer (Desktop Icon with a circle)
- Press on the + icon near SSH
- A prompt opens up, enter in the prompt

  ```
  ssh sss@172.16.223.168
  ```

- The prompt then asks you for your SSH configuration file to update, press on the first option
- After doing all of these steps, you have successfully created the connection string to the VM server, but you need to activate it now
- To activate it, press on the right arrow, or right clicking that says “Connect to Current Window”
- A prompt opens up, that asks you to choose a platform of the remote host
- Choose Linux as the server is on Linux platform
- The prompt then asks for the password, enter the password thats on the VM
- To check if you have connected successfully, you should see “SSH: 172.16.233.128” in the bottom left of your screen

To view the folders on the VM, go to explorer on visual studio code, and press on open folder to connect to the remote, you should see “home/sss/”, press enter and then enter the password that is with the VM to view the contents of the server.

To execute a php file, you use something like this:

```
php helloworld.php
```

Make sure to use **cd** to go to the directory where you are saving the php files, and you can also use **ls** or **ls -l** to view all the contents inside the directory

var_dump in php displays structured information about variables/expressions including its type and value. Arrays are explored recursively with values indented to show structure. It also shows which array values and object properties are references.

```
$obj = (object) array("MCAST", "PHP", "PROGRAMMING");
var_dump($obj);
```

print_r() displays information about a variable in a way thats readable by humans. Array values will be presented in a format that shows keys and elements similar notation is used for objects. For example:

```
$obj = (object) array("MCAST", "PHP", "PROGRAMMING");
print_r($obj);
```

14/10/2025

Last edited time	October 28, 2025 11:13 AM
Subject	Server Scripting

Lesson Notes

When creating a function in php, you can include this line of code to allow strict requirements of data types:

```
<?php declare(strict_types = 1); //strict requirement
```

28/10/2025

Last edited time	November 20, 2025 12:23 PM
Subject	Server Scripting

Lesson Notes

Laravel is PHP based framework that has a lot of built in functionalites where you won’t need to reinvent the wheel.

Laravel follows the MVC pattern

Laravel is already installed properly on the virtual machine, but there is MCAST notes for the installation of laravel

To check if you’re in the home directoy, you can use this command:

```
pwd
```

To create a laravel project in the home directory, you can use this command:

```
composer create-project laravel/laravel laravelproj
```

laravelproj in this case is the name you wish to use

Since there is only one version of laravel installed, you don’t have to specify the version

(To be safe, still specify a version in the case that some files would be missing, you can include the version by typing laravel/laravel=10 for Laravel v10)

After hitting enter, the project gets fully created, you can find it in the directory.

04/11/2025

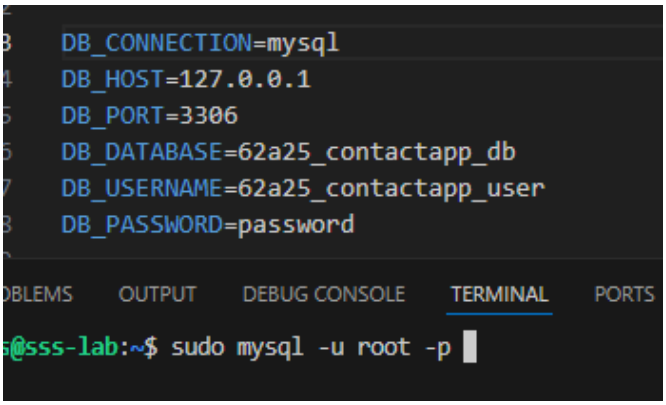
Last edited time	October 17, 2026 10:12 PM
Subject	Server Scripting

Lesson Notes

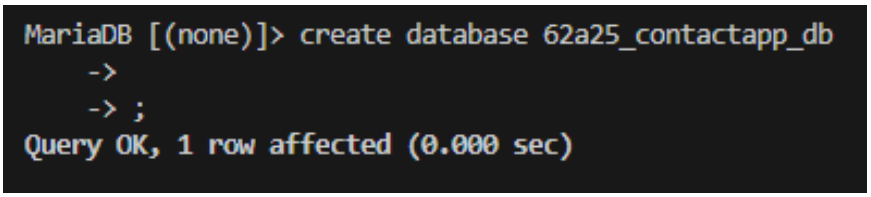
Steps to create a database for a laravel project

How to set up database and user through my sql:

- Replace the db_database, db_uername and db_password, then enter mysql



- Enter password for sss, then database password




```
    -> ;
Query OK, 1 row affected (0.000 sec)
```

```
MariaDB [(none)]> create user '62a25_contactapp_user'@'%' identified  by 'password';
Query OK, 0 rows affected (0.002 sec)
```

- So basically the @ sign means the hostname where the user is accessing the database. The % symbol is a wildcard so it's gives access from all hostnames

```
MariaDB [(none)]> grant all on 6a25_contractapp_db.* to '62a25_contactapp_user'@'%';
Query OK, 0 rows affected (0.000 sec)
```

```
MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.000 sec)
```

```
exit
```

- You can use the command to exit the database and go back to using the normal command prompt.

```
php artisan migrate
```

- Migration files is code that is generated, and this code explains to laravel the relationship and tables created on the database and to update it accordingly.

Create/add a new MySQL user account

MySQL defines users with a username and the hostname or IP address that they're using to access the MySQL instance. To create a new user in MySQL, specify the username, the hostname the user can use to access the database management system, and a secure password. This process is essential for creating new user accounts and managing access:

```
mysql> CREATE USER
'local_user'@'localhost'
IDENTIFIED BY   'password';
```

This command will allow the user with username *local_user* to access the MySQL instance from the local machine (localhost) and prevent the user from accessing it directly from any other machine. Alternatively, you can use a wildcard character (%) in the host definition to grant access to the MySQL instance for a user:

```
mysql> CREATE USER
'subnet_user'@'10.0.%'
IDENTIFIED BY   'password';
```

In the above example, the `10.0.%` specifies that the user can access the MySQL instance from any client that has an IP address beginning with `10.0.`. You may use the wild card at any level of the IP address in the host definition.

10/11/2025

Last edited time	November 11, 2025 12:22 PM
Subject	Server Scripting

Lesson Notes

To create a migration, use this command:

```
php artisan make:migration create_companies
```

To use a database in mariadb:

```
use home_contactapp_db
```

To check table content in maria db:

```
DESCRIBE companies;
```

To create a model on laravel, use this:

```
php artisan make:model Company
```

To interact with a model and test before actually coding the model on laravel, use this:

```
php artisan tinker
```

To use the model in artisan tinker, use this:

```
use App\Models\Company
```

This is an example of ORM (Object Oriented Model).

To create rows in artisan tinker, do this:

```
$company = new Company()

$company->name = "MCAST" //Display the same value if successf
ul after hitting enter
$company->address = "Paola"
$company->website = "https://mcast.edu.mt"
$company->email = "help@mcast.edu.mt"

$company->email //Showcases the value entered in the Object

$company->save() //If it displays true, then a row was create
d successfully
```

To select a specific record by id, use this (Make sure you are still using the model):

```
Company::find(1)
```

To use a where clause in tinker, use this:

```
Company::where('address', 'Paola')
```

To get all records in table, use this:

```
Company::get()
```

To enable mass assignment, create variable this in the PHP model code:

```
protected $fillable = ['name', 'address', 'website', 'emai
l'];
```

To use this mass assignment in artisan tinker

```
$companyBettson = New Company()
$companyBettson = Company::create(['name'=>'Bettson', 'addres
s'=>'Ta Xbiex', 'website'=>'bettson.com', 'email'=>'info@bett
son.com'])
```

11/11/2025

Last edited time	October 17, 2026 10:14 PM
Subject	Server Scripting

Lesson Notes

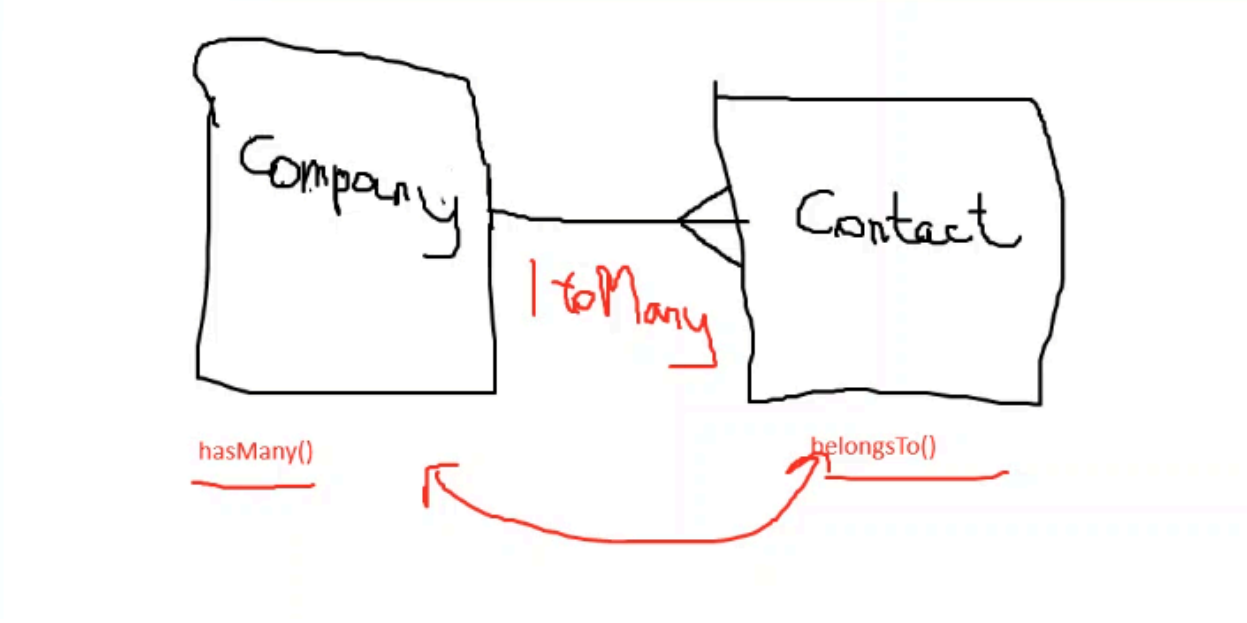
This is to put the contents of a table into an array

```
Company::all()->toArray()
```

To delete a record:

```
Company::delete(1) //Could be wrong, might not need to use
```

Relationship Concept for Models



WHENEVER YOU MAKE CHANGES TO A MODEL, EXIT ARTISAN TINKER AND USE IT AGAIN

```
> use App\Models\Company
> use App\Models\Contact
> Contact::get()
= Illuminate\Database\Eloquent\Collection {#5942
    all: [
      App\Models\Contact {#5944
        id: 1,
        first_name: "Dale",
        last_name: "Fenech",
        phone: null,
        email: "fuckyou@kissmyass.com",
        address: "Pluto",
        company_id: 1,
        created_at: "2025-11-11 11:00:16",
        updated_at: "2025-11-11 11:00:16",
      },
    ],
  }

> $contact2 = new Contact
= App\Models\Contact {#5957}

> $contact2->first_name = "Gojo"
= "Gojo"

> $contact2->last_name = "Satoru"
= "Satoru"

> $contact2->address = "Japan"
= "Japan"
```

```
> $contact2->email = "email@.com"
= "email@.com"

> $company = Company::first()
= App\Models\Company {#5250 …7}

> $company->contacts()->save($contact2)
= App\Models\Contact {#5957 …8}

> $contact2
= App\Models\Contact {#5957
    first_name: "Gojo",
    last_name: "Satoru",
    address: "Japan",
    email: "email@.com",
    company_id: 1,
    updated_at: "2025-11-11 11:14:24",
    created_at: "2025-11-11 11:14:24",
    id: 2,
  }

  OR

  $contact2->company_id = $company->id
  $contact2->save()
```

```
$contact3 = ["first_name"=>"Joe", "last_name"=>"Borg", "emai
l"=>"joeborg@gmail.com", "address"=>"Malta"]

> $contact3
= [
```

```
    "first_name" => "Joe",
    "last_name" => "Borg",
    "email" => "joeborg@gmail.com",
    "address" => "Malta",
  ]

> $company->contacts()->create($contact3)
= App\Models\Contact {#5967 …8}
```

18/11/2025

Last edited time	November 18, 2025 11:38 AM
Subject	Server Scripting

Lesson Notes

To run the web laravel application, run this command:

```
php artisan serve --host 0.0.0.0 --port 8000
```

If you want to see all routes, or all routes in a specific path, use these commands:

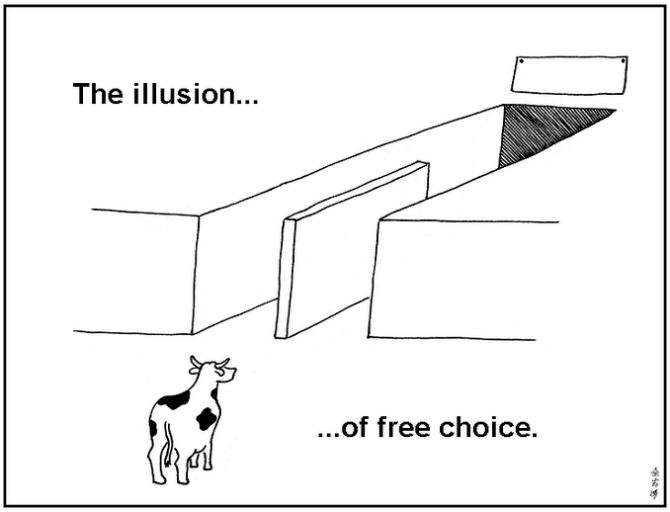
```
php artisan route:list
php artisan route:list --path=contacts
```

(CHECK GITHUB FOR MORE NOTES)

https://github.com/stackoverlearn/SSS-6.2A_25

To create a controller, use this command:

```
php artisan make:controller ContactController
```


The illusion...
...of free choice.

01/12/2025

Last edited time	December 1, 2025 2:26 PM
Subject	Server Scripting

To get a zip file to put in a project in laravel, you can use this command:

```
wget [Zip file link, you can get this from a git project by c
opying the link address of the download zip button]

         [In this case: https://github.com/stackoverlearn/boo
tstrap-crud-template/archive/refs/heads/master.zip]
```

To unzip the file, use this command:

```
unzip [blank]
            [In this case, master.zip]
```

When handling js or css files, make sure they are placed inside the public folder of a project, and to use laravel's relative path code line, which is this:

```
<link href="{{ asset('css/bootstrap.min.css') }}" rel="styles
heet">
<link href="{{ asset('css/custom.css') }}" rel="stylesheet">

...

<script src="{{ asset('js/jquery.min.js') }}"></script>
```

```
<script src="{{ asset('js/popper.min.js') }}"></script>
<script src="{{ asset('js/bootstrap.min.js') }}"></script>
```

INSTALL EXTENSIONS TO MAKE THIS EASIER

05/01/2026

Last edited time	January 10, 2026 5:28 PM
Subject	Server Scripting

`php artisan migrate:refresh` → Use this command to refresh the migrations if you made some changes, make sure to deleate existing tables as well.